Junseong Lee

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Education

University of Illinois Urbana-Champaign Expected May 2027
B.S. in Finance and Data Science | Minor in Computer Science *GPA: 3.77 / 4.00*

Relevant Coursework: Linear Algebra, Discrete Math, Data Structures, Probability and Statistics, Calculus, Machine Learning, Database Systems, Data Science, AI Ethics

Publications

Beyond explainability: a multi-dimensional framework for behavioral acceptability and role separation in medical AI
Junseong Lee *et al.*
Manuscript in preparation

Preserving independent judgment and verification beyond de-skilling in clinical and agentic AI systems
Yoojin Nam, Jeongsoo Yoon, Hee Sang Oh, Dong Yeong Kim, ***Junseong Lee*** *et al.*
npj Digital Medicine (under revision)

Expertise modulates automation bias and sentinel behavior in human-AI collaborative diagnosis of neonatal pneumoperitoneum
Junseong Lee *et al.*
npj Digital Medicine, DOI:10.1038/s41746-026-03399-3

Automated landmark detection and view positioning assessment of shoulder Grashey view radiographs using cascade deep learning: A dual-center validation study
Seung Min Ryu, Changhyun Park, ***Junseong Lee*** *et al.*
Medical Physics, DOI:10.1002/mp.70285

Presentations

Preserving evidence, context, and disagreement in mechanistic microbe-host graph QA
Junseong Lee *et al.*
NeurIPS 2026, GenAI4Health Workshop — Accepted for Poster Presentation

Bounded verification and asymmetric recoverability in biomedical knowledge-graph construction
Junseong Lee *et al.*
NeurIPS 2026, AI for Science Workshop — Accepted for Poster Presentation

When a good AUC misleads: quantifying physician-AI interaction with optimized and biased models in pediatric radiology
Junseong Lee *et al.*
IPR Congress 2026, Poster Presentation (Abdomen; presented by Hee Mang Yoon)

Representation-focused batch sampling enhances performance of efficiently adapted foundation model for detecting neonatal pneumoperitoneum on cross-table lateral radiographs
Junseong Lee *et al.*
RSNA 2025, Poster Presentation (Imaging Informatics)

Experience

Research Intern May 2026 – Aug 2026
Massachusetts General Hospital and Harvard Medical School (PI: Synho Do) *Boston, MA*

- Built **BiomeTrail**, a biomedical evidence graph that preserves molecular intermediates, source sentences, study-context metadata and opposing claims across microbe–host relationships.
- Built a cost-optimized two-stage pipeline: a deterministic NER + entity-grounding stage (**2.5M**-surface automaton) reducing 240K papers to an **8.5M** candidate-relation pool, then targeted LLM relation extraction (AWS Bedrock) verifying each edge against its source sentence—served via Neo4j and an interactive 3D graph explorer (React).
- Enforced a type-safe ontology schema and LLM-based grounding validation under a measurement-first workflow—independent precision audits and held-out gold sets gate each pipeline stage before paid scale-up.

AI Research Engineer Intern Feb 2025 – Feb 2026
Asan Medical Center (PI: Namkug Kim) *Seoul, South Korea*

- Developed and benchmarked a RAD-DINO model adapted with LoRA for neonatal pneumoperitoneum detection, achieving an AUROC of 0.948 on an external set of 125 radiographs from 11 hospitals.
- Co-designed a 14-reader study comparing unaided, reliable-AI and error-injected-AI conditions; analyzed 1,750 reader-case decisions using mixed-effects models and first-authored the manuscript, published in npj Digital Medicine.
- First-authored a narrative review examining how explanation, uncertainty, intervention timing and role allocation shape the use of clinical AI; preparing the manuscript for resubmission.

- Developed a web-based educational sandbox for practicing the identification and correction of erroneous AI suggestions.
- Engineered a representation-focused batch sampler for the vision-transformer classifier that improved learning of rare disease patterns, increasing sensitivity by **23%**.
- Developed a multimodal survival model for post-transplant liver cancer (HCC) under competing risks by fusing 3D CT with clinical variables (C-index **0.74**); explored tradeoff in parametric vs. non-parametric approaches and analyzed multimodal explainability.

Squad Leader and Software Engineer Oct 2023 – Apr 2025
8th United States Army, KATUSA *Camp Humphreys, South Korea*

- **Awarded Army Commendation Medal** for building and deploying an event-management web app (Python, Flask, OAuth2, Google API, AWS EC2, Nginx, Gunicorn), processing **800**+ participants across **32 events**, cutting manual reconciliation by **75%** and data-entry errors by **95%**.
- Provided network and systems support for the technology unit during field exercises; managed 12 soldiers and served as a liaison between U.S. and Korean command during joint trainings.

Projects

Gaze-VQA: Multi-View Spatial Reasoning Benchmark Dec 2025 – May 2026
MONET Lab (PI: Klara Nahrstedt)

- Designed a benchmark to probe vision-language models' human-gaze understanding across four tasks: *Gaze Target Recognition*, *Relative Orientation Reasoning*, *Cross-View Visibility Estimation*, and *Viewpoint-Based Accessibility*.
- Generated the benchmark dataset from the MVGT dataset and built the evaluation harness, combining standard text metrics with an **LLM-as-a-Judge** to score semantic correctness beyond exact match.

4th Place: MICCAI 2025 3D-VLM Challenge — Radiology Report Generation Jun – Sept 2025
Team 'Pneumo-Lingo'

- Placed **4th** of 160+ international teams by adapting BIMBA (CVPR 2025) with a segmentation-mask-aware interleaving of spatiotemporal tokens for 3D CT–text fusion. *(Work later extended into an ECCV 2026 paper, arXiv:2506.23102; acknowledged contributor.)*
- Conducted exploratory data analysis of the challenge dataset and constructed Llama-8B tailored instruction sets to guide fine-tuning.
- Built a LangChain pipeline (GPT-4o mini) to structure and classify 6 organ-level report inputs via chain-of-thought reasoning into clinically relevant labels.

Recognitions

Fred S. Bailey Scholarship 2026
$3,000 for academic merit and leadership potential *YMCA*

Woojin Scholarship 2026
National recipient ($1,500) for excellence in STEM and community service *KSEA*

1st Place: Senior Capstone 2026
Most voted outstanding purpose project *BUS401*

1st Place: Leelyn Smith Wealth Management Competition 2026
$500 for outstanding performance in wealth management planning *FIN435*

Army Commendation Medal 2025
For proactive efforts to streamline battalion workflow *2nd Infantry Division, 8th U.S. Army*

Army Achievement Medal 2024
Outstanding performance, Freedom Shield field exercise *2nd Infantry Division, 8th U.S. Army*

Cozad New Ventures AWARE Award 2023
Awarded $1,250 in funding for startup 'Scroll' *UIUC*

Key Skills

Languages: Python, C++, SQL, Java, Bash, Git
AI/ML: PyTorch, CUDA, LangChain, MONAI, scikit-learn, HuggingFace, OpenCV, Ollama
Tools & Platforms: Docker, Slurm, AWS, Neo4j, Linux, Cloudflare